Mateligent iDEAS EAP v2.1 Flex Sensor Communication Requirements:

AX-008-29, AX-008-30, AX-008-39 firmware versions

AX-008-55 (high speed) firmware version

FS – Flex Sensor

RB – Radio Board (900MHz radio board)

AD – Another Device (hardwired interface)

General:

One or more **FS**s are connected to a 900MHz radio board (**RB**), or other hardwired device.

The **FS**s are based on a Cypress CY8C4xxx PSoC4 processor, connected to an EAP element, providing two readings, an EAP measurement and a temperature measurement. Calibrated sensors return an EAP measurement in % strain. Uncalibrated sensors return raw EAP measurement count values.

Data transmitted from **FS** may either be in:

1. Binary format (most power efficient)
2. Character format (ASCII). This has larger data packets and longer data transmission times and thus higher power consumption, but data is human readable via a terminal program on a PC or other device.

Connection Interface:

The **FS** has 4 pins to interface to the 900MHz **RB**.

1. Power – 2.6 to 3.3Vdc battery power from the **RB.** 3.0Vdc nominal.
2. Ground
3. TX – TTL level asynchronous serial data to the **RB** from the **FS**
4. RX – Wake/poll signal or TTL level asynchronous serial data from the **RB** to the **FS.**

Power management:

1. The **RB** will have its own power management.
2. The **FS** will stay in low or ultra-low power mode as much of the time as possible, except when awakened by the RX signal from the **RB** to take a reading (non-streaming mode), or when it is streaming a reading.

FS Modes/Use cases:

1. RF Operation (**FS** connected to a 900MHz **RB** or other ultra-low power device)
    a. Idle – in ultra-low power mode, waiting for wake up from **RB**.
    a. Measurement mode – entered when awakened by RX wake/poll signal from **RB**.

   i. Power up
   ii. Measure sensor
   iii. Transmit measurement data on TX line (see Binary data format)
   iv. Go back to Idle

2. Hardwire Operation (**FS** connected to another device (**AD**), hardwired, low power but not ultra-low power consumption)
   a. Idle – in ultra-low power mode, waiting for wake up from **AD**.
   b. Measurement mode – entered when awakened by RX wake/poll signal from **AD**.
      i. Power up
      ii. Measure sensor
      iii. Transmit measurement data on TX line (see Character data format)
      iv. Go back to Idle

3. Streaming Operation (**FS** connected to another device (**AD**), hardwired, without low power consumption constraints)
   a. Idle – in low power mode, with internal measurement timer running.
   b. Measurement mode – entered when measurement timer expires.
      i. Power up
      ii. Measure sensor
      iii. Transmit measurement data on TX line (see Character data format)
      iv. Go back to Idle

4. Calibration/Development mode
   a. For more details on command set, see “Commands” section of this document.
   b. Automatically enters Calibration/Development mode if Span and Zero Calibration data is blank.
   c. At measurement rate, measure EAP and temperature. Transmit data in character format.
   d. Default measurement rate 10Hz
   e. Measurement rate set and read with command “*CP”*
   f. Calibrate: With EAP relaxed, send zero calibrate command “*CZ=1*” **FS** stores zero calibration counts. Zero calibration value stored in non-volatile (flash) memory.
   g. Calibrate: Stretch element to fixed location (66.6% of nominal extension), send span calibrate command “*CS=1*”, **FS** stores span count values. Span calibration value stored in non-volatile (flash) memory.
   h. **FS** uses zero calibration and span calibration values to convert raw measurement counts to a strain value in %. Conversion is linear, no curve fitting is done at this time.
   i. Temperature calibration – none. Cypress calibration of +/-1 °C (typ), +/-5 °C (max) over -40°C to +85°C shall be acceptable.
   j. Temperature compensation: x measurement counts/°K. Where x is calculated as x = 0.001261791 * CalZero - 0.3682680. Temperature Is the difference between the measured die temperature from the calibration temperature stored in the microcontroller.
   k. Sensor serial number shall be a 32 bit hexadecimal value (This needs to be validated, and sufficient program memory space needs to be available to convert and transmit it –

TBD. Are there any other ideas for serial number formats?). Two integer numeric entry commands shall be used to set the hexadecimal serial number in non-volatile (flash) memory, "*S1=57005*" for the high 16 bits, and "*S2=48879*" for the low 16 bits.

l. Command to read serial number "*SN?*" will return "*DEADBEEF*" for the above example.
m. Command to set/read nominal sensor length "*SL=nnn*" where *nnn* = sensor length in integer mm.
n. Command to set drive strength for higher capacitance sensors "*CI=yyy*". Future feature, implementation date TBD.

5. Analog/PWM output mode – enabled if: CA=1, CD=1, CP <= 100, and sensor is calibrated. 120% strain = 100% analog output. Raw output signal is a PWM waveform (32kHz base frequency) that can be buffered and filtered to produce an analog voltage. See AX-008-26 terminal block board for an analog out schematic example.

6. Re-flash **FS** firmware (bootloader)
   a. Bootloader stored in Cypress controller non-volatile memory.
   b. Preferred bootloader entry method: Initiate bootloader with a serial command "SF=1".
   c. Bootloader mode is indicated by a slowly flashing blue LED.
   d. Reprogram flash with new application firmware image via serial Rx and Tx pins.
   e. A PC program (Cypress Bootloader Host or custom bootloader utility) and FTDI TTL-232R-3V3 cable will be needed.
   f. Alternate bootloader entry method – connect PCB bootload input trace to ground prior to powering up sensor. At power up, the processor checks the PCB bootload input state, and enters the bootloader mode if it is grounded

7. Production Test – TBD

Measurement Mode data formats:

1. Binary data format, **calibrated** sensor. (Uncalibrated sensor defaults to character data format.) Communication settings: 9600 baud, 1 start bit, 8 data bits, 1 stop bit. NOTE: baud rate for AX-008-55 (high speed) version is 115,200 baud.
   a. 2 bytes temperature in degrees (K) in tenths. 273.1 °K = 0.0 °C
   b. 2 bytes primary reading in percentage in hundredths.
      i. (0x0000 = 000.00% to 0x7FFF = 327.67%)
      ii. Expected maximum value approx. 120.00%
      iii. Maximum value saturated to 200.00%
      iv. Expected minimum value – saturated to 0.00%
      v. Error message for disconnected sensor, overflow, etc (0x8000/-32768?)
   c. 1 byte status flag – bit 0x01 set for sensor error, bit 0x20 set for saturated (calibrated) measurement.
   d. 1 byte checksum (sum of previous bytes in message)

2. Character (ASCII) data format, **calibrated** sensor. Communication settings: 9600 baud, 1 start bit, 8 data bits, 1 stop bit. NOTE: baud rate for AX-008-55 (high speed) version is 115,200 baud.
   a. Message string format: "000.00%,000.0K"

b. 6 bytes primary reading in percentage in hundredths with decimal point.
    i. ("000.00" = 000.00% to "327.67" = 327.67%)
    ii. Expected maximum value approx. 120.00%
c. 1 byte "%", indicating calibrated percentage reading.
d. 1 byte comma
e. 5 bytes temperature in Kelvin degrees in tenths with decimal point. ("273.1" = 273.1K)
f. 1 byte "K", indicating Kelvin temperature units.
g. Message string format if sensor measurement error present (measurement overflow or open circuit): "EAP error"

3. Character (ASCII) data format, **uncalibrated** sensor. Communication settings: 9600 baud, 1 start bit, 8 data bits, 1 stop bit. NOTE: baud rate for AX-008-55 (high speed) version is 115,200 baud.
    a. Message string format: "00000,000.0K"
    b. 5 bytes primary reading – raw measurement counts
        i. Minimum value 0
        ii. Maximum value 65535
    c. 1 byte comma
    d. 5 bytes temperature in Kelvin degrees in tenths with decimal point. ("273.1" = 273.1K)
    e. 1 byte "K", indicating Kelvin temperature units.
    f. Message string format if sensor measurement error present (measurement overflow or open circuit): "EAP error"

Commands:

1. Commands consist of three characters: i.e. "*CL?*" Or "*CL=*"
2. The first character is the type of data: "*C*" is for Calibration/Configuration data, "*S*" is for Sensor/Serial number data.
3. The second character further specifies the data being referenced.
    a. For data where the first character is "*C*" the second character defines the following specific data:
        i. "*A*" = ***C***onfigure ***A***nalog/PWM output. 1 = analog/PWM output is enabled if CD=1, CP not equal to zero, (and CP <= 100 for AX-008-30), and sensor is calibrated. Default: *CA=0*.
        ii. "*B*" = ***C***alibration ***B***urst measurement (requires parameter between 0 and 7 inclusive). After receiving a "*CB=n*" command the FS will take a series of 2^n measurements and transmit the averaged result(s). This facilitates an application processor implementing a non-standard or more complex calibration after commanding the **FS** to take calibration measurements at certain custom operating points. The samples are stored in the ring buffer memory, so this command can be used to seed or reset the ring buffering ("*SA=4*" thru 7).
        iii. "*C*" = ***C***alibration temperature ***C***oefficient (requires parameter between 0 and 65535) If not 0 or 65535, the parameter value sets custom Calibration temperature Coefficient in: counts * -1024 per °K). Default: *CC*=0xffff, blank. Requires *CZ*, *CS*, and *CT* parameters to be set before it will compensate the measured values.

iv. "*D*" = ***C***haracter ***D***ata (0 = data in binary form, 1 = data in character form). Default: *CD=1*. (Note: this parameter is not stored in non-volatile flash memory until the "*CR=1*" ***C***onfiguration ***R***ecord command is received.)

v. "*L*" = ***C***onfigure ***L***ED (possible LED configurations: 0 = no LED, 1 = brief green blink at measurement. Additional are possible). Default: *CL=1*.

vi. "*I*" = ***C***alibration Current (***I***). EAP measurement current setting in 300nA integer increments. Default: *CI=4.* (Note: this parameter is not stored in non-volatile flash memory until the "CR=1" **C**onfiguration **R**ecord command is received.)

vii. "*P*" = ***C***onfigure ***P***olling Rate. Default: *CP=10*.

1. 0 = no polling. Wake on RX wake/poll signal for measurement only. NOTE: AX-008-55 (high speed) version at 115,200 baud must have a dummy character sent to wake it up if CP >1000 or CP=0.
2. Non-zero polling rate = number of 100Hz polling ticks to transmit data (Max 10Hz polling implemented at this time, non-zero values between 1 and 9 will be saturated to 10 (10Hz) minimum polling rate. NOTE: AX-008-55 high speed version uses 1mS polling interval. Minimum = 3 (333Hz, must use binary data mode), default = 100 (10Hz). Values of 1 or 2 will be saturated to 3.)
3. Polling rate = 65535 => 10.923 minute maximum polling period (1.09 minute maximum for AX-008-55 high speed)
4. Polling rate timer accuracy TBD – can implement internal trimming based on internal timer and temperature measurement if there is sufficient program memory. +/- 5% may be possible with frequent trimming (power budget permitting as well).
5. Note: this parameter is not stored in non-volatile flash memory until the "*CR=1*" ***C***onfiguration ***R***ecord command is received.

viii. "*R*" = ***C***onfiguration ***R***ecord. Store the configuration data in non-volatile flash memory. Wait a minimum of 50mS for this non-volatile memory update.

ix. *"S"* = ***C***alibration ***S***pan. Send "*CS=1*" to start a span calibration event. Calibration point = 2/3rds nominal span. Default: 0xffff, blank. Returns "CS=Error" if zero calibration value is blank, Zero Calibration needs to be set first. Send "CS=nnnnn" to set the Calibration Span to a desired value nnnnn. This value must be greater than 100.

x. "*T*" = ***C***alibration ***T***emp. Enter/read temperature (in °K) that calibration was performed at. Enter temperature in tenths **without** typing the decimal point. For example, enter "2981" for 298.1°K/25.0°C. Default: 0xffff, blank. If calibration temperature value is blank or 0, temperature compensation of the measured values is disabled. If CC parameter is blank or 0, then the temperature compensation slope (Y) is dependent on the *CZ* value as follows: Y = (0.001261791 * *CZ)* – 0.3682680, Y is in counts/°K. The difference between the current measured temperature and the entered calibration temperature is stored and used as an offset calibration for the measured temperature.

xi. "*Z*" = ***C***alibration ***Z***ero. Send "*CZ=1*" to start a zero calibration event. Must be performed before Calibration Span command. Default: 0xffff, blank. Send "CZ=nnnnn" to set the Calibration Zero to a desired value nnnnn. This value must be greater than 100.

b. For data where the first character is "*S*" the second character defines the following specific data:
   i. "*1*" = ***S***erial number word (16 bit value) ***1***. Data entry ("*S1=xxxxx*") only.
   ii. "*2*" = ***S***erial number word (16 bit value) ***2*** Data entry ("*S2=yyyyy*") only. NOTE: S1/S2 are not re-writeable after power cycle. To erase, issue Calibration Erase ("*CE=1*") command to erase serial number after power cycle. Sensor recalibration will also be required.
   iii. "*A*" = ***S***ensor ***A***veraging. The SA parameter is a power of 2 exponent. Default is "*SA=0*" (2^0 = 1), one un-averaged sample per reading. For "*SA=1*" to *"SA=3",* 2^SA samples are rapidly taken (oversampled) and averaged. For example, "*SA=3*" → 2^3=8 samples are taken and averaged. For "*SA=4*" to *"SA=7",* only one sample is taken. It is put into a ring buffer, and 2^SA previous samples are averaged. Query "*SA?*" returns exponent value. Valid "*SA=*" parameters are 0 (no averaging), 1 (2 oversampled samples averaged), 2 (4 oversampled samples averaged), 3 (8 oversampled samples averaged), 4 (one sample measured, reported value is this measurement averaged with 15 previous samples), 5 (one sample measured, reported value is this measurement averaged with 31 previous samples), 6 (one sample measured, reported value is this measurement averaged with 63 previous samples), and 7 (one sample measured, reported value is this measurement averaged with 127 previous samples). (Note: this parameter is not stored in non-volatile flash memory until the "*CR=1*" ***C***onfiguration ***R***ecord command is received.)
   iv. "*F*" = ***S***ensor ***F***irmware. Query "*SF?*" returns software version as a string (character data mode) or 32 bit binary value (binary data mode). Command "*SF=1*" to enter bootloader mode and upload a new firrmware version.
   v. "*L* = ***S***ensor ***L***ength – length of sensor element in integer mm. NOTE: not re-writeable after power cycle. To erase, issue Calibration Erase ("*CE=1*") command to erase sensor length after power cycle. Sensor recalibration will also b required. Default: 0xffff, blank.
   vi. "*N*" = ***S***erial ***N***umber. Query only. "*SN?*" To read whole serial number. Default: 0xffff, blank.
   vii. "S" = ***S***ensor ***S***leep – Upon receipt of "*SS=1*", A **FS** that is in Streaming Operation mode will acknowledges the command by sending "Sleep", completes its current measurement cycle, transmits the data, and then goes to deep sleep. To wake up **FS**, **AD** transmits a dummy character, which is not parsed by the **FS**.

4. The third character is either a "*?*" or an "*=*".
   a. "*?*" indicates the command is in query of a particular value.
   b. "=" indicates the command sets a value, in which case the new value should follow the "=" character. All entered values are 16 bit unsigned quantities (0 to 65535) only. Send a carriage return character to terminate numeric entry.
   c. Responses to a "*nn?*" query have an additional carriage return/line feed to insert a blank line after the query response. This helps distinguish query responses from correct command set "*mm=xxxxx*" echo responses.

d. Characters are parsed immediately one-by-one as they are received. Any illegal characters shall reset the parsing algorithm. Command re-entry must restart from the beginning of a command sequence (no backspace support).

5. Examples:
    a. “*CZ=1*” to perform a zero calibration. “*CZ?*” to request the zero calibration data.
    “*CS=1*” to perform a span calibration. “*CS?*” to request the span calibration data.
    b. “*CL = 0*” Set LED configuration to “0”.
    c. “*CL = 1*” Set LED configuration to “1” (Brief green LED blink at measurement)
    d. “*S1=57005*” for the high 16 bits of the serial number, and “*S2=48879*” for the low 16 bits.
    e. “*SL=100*” to save a sensor length of 100mm. “*SL?*” to request the sensor length data.
    f. “*SN?*” will return a hexadecimal serial number of “*DEADBEEF*” for the above example.
    g. “*CP=0*” and “*CD=0*” set the **FS** to RF operation mode.
    h. “*CP=2000*” sets the **FS** to streaming mode, with a measurement interval of 20 seconds.

6. Command parsing will be blocked by any measurements in progress. Thus, at high measurement duty cycles, it can be helpful to increase the polling rate value to 10 or greater to reduce the measurement blocking time. In this case, once all the other settings are configured, set the polling rate to the desired value last.
7. Following the successful parsing of a command, the command is executed and then the response string is transmitted confirming the setting. Then after this transmission, the parameter is stored in non-volaitle memory (except for “CL=”, “CI=”, “CM=”, “CP=”, and “SA=” commands). After the confirmation transmission, allow minimum of 50mS for this non-volatile memory update. “*CR=1*” and “*CE=1*” commands also reconfigure other operational parameters. For these two commands, allow a total of 150mS after the confirmation transmission before sending another command or expecting measurements to resume.
8. During the command execution, confirmation transmission, and non-volatile memory write, all measurement activity is blocked until the command process is complete.
9. If the sensor appears to not be responding to commands, enter several characters that are not part of the command set to ensure the parsing algorithm is reset. Then you can enter a simple data reporting command to verify a correct response. For example, enter several “a” characters to reset the parsing, followed by “SF?” to see the Sensor Firmware string and see that the sensor is responding. Then enter your desired command.
10. If the combination of sensor oversampling, measurement time, number of active sensors, and data transmit time exceed the specified polling rate the sensor sends a “Measurement Period Overflow” message following the measurement data.

Suggested RF mode calibration procedure:

1. Follow bootloader programming procedure to update EAP strip firmware to latest version.
2. Connect EAP strip to PC running a terminal program such as TerraTerm or PuTTy.
3. Power up the EAP strip.

4. Initial configuration with blank zero and span calibration is to stream raw measurement counts as character data at a 10Hz measurement rate. Verify the reported count values are as expected.
5. With the sensor at nominal calibration temperature and EAP zero extension, send a *"CZ=1"* command to perform a zero calibration. The sensor averages 16 measurements and reports the stored zero calibration value.
6. Extend the sensor to 66.66% (2/3rds) of nominal extension. Send a "CS=1" command to perform the span calibration. The sensor averages 16 measurements at this extension, multiplies them by 3/2 and reports the stored span calibration value.
7. The sensor now reports extension as a percentage.
8. The zero and span calibration values can be requested from the sensor with "CZ?" and "CS?" commands. Verify desired sensor response/accuracy/resolution.
9. Use the "*CD=x*", "*CP=yyy*", "*CL=z*" commands to configure the desired behavior.
10. Enter the serial number, sensor length, and calibration temperature if desired.

Changelog:

11/2/2016 – Initial v2.1 version, forked from EAP v2.0 / Firmware AX-008-08_RevBp3. No IMO oscillator trim, 1 second character mode delay for FTDI/windows serial mouse bug. Target: CY8C4146AZI-S433. Firmware version AX-008-13_RevAp1.

11/4/2016 – Implemented IMO oscillator trim, improving accuracy from ±2% to ±0.2%. Increased character mode delay to 2 seconds. Target: CY8C4146AZI-S433. Firmware version AX-008-13_RevAp2.

11/16/2016 – Changed current sources to both be 300nA per bit, default EAP drive = 4 (for same 1.2uA). Added ring buffer sampling mode for SA=4 (average of last 16 samples) to SA=7 (average of last 128 samples). Oversampling only for SA=1, SA=2, SA=3. Increased measurement clock from 12MHz to 24MHz for increased sensitivity. Report measurement timer overflow and keep running. Block CI=0 setting (invalid input that breaks operation). Target: CY8C4146AZI-S433. Firmware version AX-008-13_RevAp3.

11/22/2016 – Added measurement tweaks and error trapping to detect and report (in character mode) the following conditions:

- EAP init error (EAP pin won't discharge to 0V before starting measurement)
- No/low EAP error (EAP missing or capacitance too low)
- EAP grounded error (EAP pins shorted together or to ground)
- EAP overflow error (EAP capacitance not charging to hi threshold before timeout)

Added temperature sensor offset calibration via "CT=xxxx" command. When valid calibration temperature is set, the sensor takes the current temperature reading, computes the offset from the entered calibration temperature, and stores this value in non-volatile memory. This offset is then used to calibrate the measured temperature. NOTE: the temperature sensor has a native measurement resolution of 1.0 °C (1.0 °K). The offset calibration factor is rounded to the nearest integer °C (°K).

Added CP=0 measurement triggering via RX pin since v2.1 measurement electronics no longer have SW pin. To use new RX wake/poll signal to trigger measurement:

- Pulse RX pin low for 50uS – 100uS (not AX-008-55)
- Pulse RX pin low for over 1.0mS
- Send null (0x00) character to RX pin (use this for AX-008-55)
- Send 0xff character To RX pin

(Any other single character will be parsed/ignored by serial Rx routine.)

When starting in character data mode (CD= 1), sensor transmits version information, then waits for 2 seconds before starting measurement. When connected to FTDI cable, the delay avoids Microsoft Windows serial mouse USB enumeration bug. To bypass 2 second delay, send any character (character will be ignored) or pulse RX line low as described above.

Removed reference to SW pin, replaced with "RX wake/poll signal". Updated bootloader section, preferred entry method is via "*SF=1*" command.
11/22/2016 - AX-008-29_RevAp5 release
 - Fixed oversampling (SA=1-3) VrefLo comparator trigger bug that caused samples after first one to run long.
 - Fixed ring/circular buffer sample average bug (didn't average one sample).
 - Added CIRC_BUFFER compile time switch to oversample and circular buffer average for evaluation in hose application.

There is a 'regular' RevAp5 version in the zip, plus 5 sets that have "CB=n", where n = 1 to 5, and is the size of the circular buffer in 2^n samples. These CB versions also have SA=3 for 8 oversamples.

1/25/2017 – AX-008-29_RevAp6 'high speed' release
 - Remove CP saturation to 10 for values between 1 and 9. This allows CP=1 for 100Hz measurements.
 - Reduce EAP discharge time from 2mS to 1mS.
 - Reduce oversampling delay between samples from 2 to 1mS.
 - Remove oversampling delay after last sample.
 - Increase UART baud rate to 38,400 baud. Character mode data packet (14 characters) transmission time = 3.7mS

Target: AX-008-20 v2.1 Hi-Res PCB assembly, processor: CY8C4146AZI-S433.

3/3/2017 – AX-008-30_RevAp6 'high speed' release
 - Ported AX-008-29_RevAp6 from AX-008-20_RevA assembly with large LQFP processor to AX-008-17_RevC thin/tiny assembly.

Target: AX-008-18_RevC v2.1 thin PCB assembly, processor: CY8C4146FNI-S433.

3/7/2017 – AX-008-29_RevAp7 and AX-008-30_RevAp7 'high speed' release
 - Fixed cypress bootloader bug. Baud rate was >10% off when using WCO crystal trim of IMO, not the ±0.2% cypress specified. Reverted to ±2% untrimmed IMO.

Target: AX-008-20 v2.1 Hi-Res PCB assembly, processor: CY8C4146AZI-S433. Firmware version AX-008-29_RevAp7.

Target: AX-008-18_RevC v2.1 thin PCB assembly, processor: CY8C4146FNI-S433. Firmware version AX-008-30_RevAp7.

3/13/2017 – AX-008-29_RevAp8 and AX-008-30_RevAp8 release – version with ring buffering and oversampling enabled for hose testing.

5/26/2017 – AX-008-29_RevAp9 and AX-008-30_RevAp9 analog release. Analog output enable if: CD=1, CP <= 100, and sensor is calibrated. 120% strain = 100% analog output.
11/15/2017 – AX-008-39-RevAp10. Initial release for -39 variant with 4125S processor. Added Calibration Zero point based temperature compensation, per Parker document "Parker EAP Sensor Gen2.1 RevE Temperature Testing Data.ppt". Merge common source files and put under version control.

12/18/2017 – NOTE: In the process of porting this code to the TI SimpleLink BLE platform, it was found that the sensor averaging buffer stores raw counts, rather than calculated percent. This leads to a pretty long lag at startup when starting cold with all 0's in the buffer. It also amplifies any non-linearity in the raw value to span calculation.

6/12/2018 – AX-008-55_RevAp11 release – High speed version optimized for 100mm sensors. Default CI = 8, CP = 100 (in 1mS ticks, or 100Hz). Note: FTDI cable/PuTTy to a PC can't keep up at CP=3 in character data mode, use binary mode for CP=3. Target: AX-008-18_RevC v2.1 thin PCB assembly, processor: CY8C4146FNI-S433.
October 12th, 2018 – Ring buffers seeded at startup with an average of 16 initial readings. The following commands no longer update the configuration data structure in non-volatile Flash memory:

i. <u>***C***</u>onfigure <u>***P***</u>olling Rate: "*CP=x*"
ii. <u>***S***</u>ensor <u>***A***</u>veraging: "*SA=x*"
iii. <u>***C***</u>onfigure <u>***L***</u>ED: "*CL=x*"
iv. <u>***C***</u>alibration Current (<u>***I***</u>): "*CI=x*"

Added <u>***S***</u>ensor <u>***S***</u>leep "*SS=1*" command. Added <u>***C***</u>onfiguration <u>***R***</u>ecord "*CR=1*" command to save the configuration data structure in non-volatile Flash memory. Added <u>***C***</u>alibration <u>***B***</u>urst measurement "*CB=n*" command, also seeds the ring buffer. Added "Measurement Period Overflow" message. "*CL=0*" now also disables LED blinks at startup.
AX-008-30_RevAp12 release – High resolution version, processor: CY8C4146FNI-S433
AX-008-39_RevAp12 release – Standard v2.1 version, processor: CY8C4125FNI-S433
AX-008-55_RevAp12 release – High speed version optimized for 100mm sensors, processor: CY8C4146FNI-S433

October 18th, 2018 – Added ‘*CC*’ Calibration temperature Coefficient command. Updated default temperature compensation behavior. Updated ‘*CT*’ description to match CZ-dependent slope calculation implemented in RevAp10. Fixed bug – would not transmit raw counts in binary mode.
AX-008-30_RevAp13 release – High resolution version, processor: CY8C4146FNI-S433
AX-008-39_RevAp13 release – Standard v2.1 version, processor: CY8C4125FNI-S433
AX-008-55_RevAp13 release – High speed version optimized for 100mm sensors, processor: CY8C4146FNI-S433

November 29th, 2018 – Added status bits reporting in binary mode for saturation and EAP measurement error. Modified checksum formula to sum all bytes in binary data message packet. Fixed ring buffer seeding error at startup. Updated ‘*CT*’ description to include temperature offset calibration.
AX-008-30_RevAp14 release – High resolution version, processor: CY8C4146FNI-S433
AX-008-39_RevAp14 release – Standard v2.1 version, processor: CY8C4125FNI-S433
AX-008-55_RevAp14 release – High speed version optimized for 100mm sensors, processor: CY8C4146FNI-S433

April 15th, 2019 – Fixed WDT overflow bug in AX-008-55 for cases with CP=0 that caused erroneous “Measurement Period Overflow” messages. Added StopWDT to non-WDT cases when sensor configuration is changed.
AX-008-30_RevAp15 release – High resolution version, processor: CY8C4146FNI-S433
AX-008-39_RevAp15 release – Standard v2.1 version, processor: CY8C4125FNI-S433
AX-008-55_RevAp15 release – High speed version optimized for 100mm sensors, processor: CY8C4146FNI-S433

October 21st, 2020 – AX-008-62 Evaluation release. Custom version based off AX-008-39_RevAp15 with 115,200 baud rate and max polling rate of 50Hz (min CP=2).

April 13th, 2022 – Added ‘*CA*’ Configure Analog/PWM output command.
AX-008-30_RevAp17 release – High resolution version, processor: CY8C4146FNI-S433
AX-008-39_RevAp17 release – Standard v2.1 version, processor: CY8C4125FNI-S433
AX-008-55_RevAp17 release – High speed version optimized for 100mm sensors, processor: CY8C4146FNI-S433